Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation

Team Number: 28

S.No	Register Number	Student Name
1	CB.SC.U4CSE23363	HASWITHA K
2	CB.SC.U4CSE23529	HASINI REDDY M
3	CB.SC.U4CSE23547	SHEELA AKSHAR SAKHI
4	CB.SC.U4CSE23761	KOUSIK SARMA LAKKARAJU
5	CB.SC.U4CSE23753	V. CHAKRAVARTHY

Guide: **DR. G R RAMYA, DR. VANDHANA S**

Department of CSE

Previous Review Comments & Action Taken

Reviewer Comments (Review 2)

- **Comment 1: Start Implementation**
  The implementation phase of the proposed federated framework must be initiated.
- **Comment 2: Working Prototype & Dashboard**
  Build a functional clinician dashboard and working prototype to demonstrate live clinical decision support.

Action Taken & Implementation Status

- **Implemented Federated Pipelines:**
  - Built decentralized PyTorch algorithms (FedAvg, FedProx, Ditto).
  - Implemented local AD-CUSUM drift detection and Client-Side Selective Personalization.
- **Built Production-Grade CDSS Workstation:**
  - **Data Warehouse**: Local SQLite clinical database.
  - **Backend API**: Uvicorn/FastAPI server running PyTorch checkpoints.
  - **Interactive UI**: Vite React medical dashboard mapping vitals charts, risk dials, and explainable attention timelines.

Core Project Inventions & Research Novelties

1. AD-CUSUM Drift Algorithm (New Invention)

Adaptive Divergence-Weighted CUSUM: Combines loss residuals, Jensen-Shannon entropy divergence (D_{JSD}), and gradient variance weighting (w_{grad}) to detect population drift without mini-batch noise false alarms.

2. DDP-ERP Personalization Algorithm (New Invention)

Dynamic Dual-Phase & Entropy Regularization: Integrates 2D temporal-vital cross-attention ($\alpha_t \otimes \beta_v$) with KL-Divergence soft-personalization, boosting test accuracy to **96.42%** (>95% target) and **0.9418 AUROC**.

3. Client-Side Selective Personalization (CSSP)

Freezes backbone weights during drift adaptation rounds, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB) and cutting adaptation time to **6 minutes**.

4. Production Clinical CDSS Workstation

Full-stack bedside decision support system linking React/Vite, FastAPI REST server, and SQLite database with 24-hour sequence hour scrubbers and SSC Hour-1 bundle checklists.

Clinical Deterioration & Phase II Completed Goals

Clinical Context:

- **High-Sparsity Time-Series:** Continuous ICU vital monitoring logs (HR, SBP, Temp, SpO_2) combined with sparse blood panel entries.
- **Privacy constraints:** HIPAA / DPDPA (Section 8) regulations mandate that private patient records cannot leave local devices.
- **Concept Drifts:** Clinical demographics vary heavily across regional ICU nodes.

SDG Alignment:

- **Primary (SDG 3):** Good Health & Well-being by detecting patient sepsis crashes early.
- **Secondary (SDG 9 & 16):** Secure and scalable digital clinical infrastructure.

Phase II Completed Milestones

- **1. Data Engineering:** Cleansed, Imputed, Standardized PhysioNet 2019 logs into active PyTorch sequence arrays.
- **2. Federated Baselines:** Implemented *FedAvg* and *FedProx* securely over PyTorch.
- **3. Personalized FPDAF Model:** Trained temporal self-attentions with localized personalization.
- **4. Clinical Workstation App:** Developed a dynamic FastAPI + SQLite + React Vite ICU dashboard overlay.

Clinical Profile of ICU Sepsis

What Sepsis is

Sepsis is a life-threatening medical emergency caused by the body's extreme, systemic immune response to an infection, which ends up damaging its own tissues, blood vessels, and vital organs.

Why it is Dangerous

If not diagnosed immediately, it rapidly progresses to Septic Shock, leading to multi-organ failure, a sudden drop in blood pressure, and high mortality rates.

The Clinical Time-Window Challenge

Sepsis is notoriously difficult to detect early because its initial symptoms resemble general infections. However, every single hour of delayed treatment increases a patient's risk of death by approximately 8%.

Role of Our Project

By continuously analyzing ICU vital signals, our FPDAF (Federated Personalized Drift-Aware Attention Framework) early-warning system acts as an early-warning radar system, alerting doctors hours before a patient clinically deteriorates, giving clinicians a vital head-start to administer antibiotics and fluids.

Sepsis Burden & Impact in India

National Mortality & Case Load

Sepsis represents a critical public health crisis in India:

- **2.9 to 3.0 Million Deaths:** Estimated annually across India due to Sepsis.
- **11.0 to 11.3 Million Cases:** Recorded annually nationwide.
- **Proportional Mortality:** Sepsis accounts for roughly **30% (3 in 10)** of all deaths in India.

Source: The George Institute for Global Health, 2020

Indian ICU Mortality Metrics

Within clinical critical care environments:

- **34% to 50%+ Mortality Rate:** Severe sepsis patients in Indian ICUs face high clinical mortality.
- **Antimicrobial Resistance (AMR):** High AMR rates complicate treatment, making hours-early detection crucial.

Source: Indian Society of Critical Care Medicine (ISCCM)

Clinical Significance & Research Objective

Given that severe sepsis mortality in Indian ICUs exceeds 34%, implementing a localized, real-time early warning system is the most critical and clinically viable strategy to save patient lives.

Proposed Framework: FPDAF

WHAT is FPDAF (Federated Personalized Drift-Aware Attention Framework)?

A privacy-preserving Clinical Decision Support System (CDSS) for Sepsis early warning. Enables collaborative, HIPAA-compliant training across decentralized hospital ICU nodes without centralizing raw records.

HOW does it work?

- **Sequence Modeling:** Extracts temporal clinical features using a shared Bidirectional LSTM.
- **Ditto Personalization:** Fine-tunes localized classifier heads (v_k) to fit hospital demographics.
- **Online AD-CUSUM Trigger:** Audits prediction loss residuals locally to detect population drift.

Core Scientific & Technical Novelties

1. Multi-Head Attention Saliency (Clinical Explainability)

Integrates a 4-head query-key temporal self-attention mechanism to output dynamic sequence saliency weights, mapping exactly which hourly vitals contributed to Sepsis alarms at the bedside.

2. Online AD-CUSUM Residual Neural Drift Monitor

Tracks local validation loss residuals round-by-round ($S_r > 3.0$). Concept drift is detected entirely client-side, avoiding constant full-parameter aggregation cycles.

3. Client-Side Selective Personalization (CSSP)

Freezes the shared BiLSTM/attention layers upon drift triggers to train local classifier heads exclusively.

- Bypasses global parameter uploads, saving **38% in communication bandwidth** (1.52 GB vs 2.48 GB).
- Cuts client-side adaptation training time to **6 minutes** (down from 25 minutes).

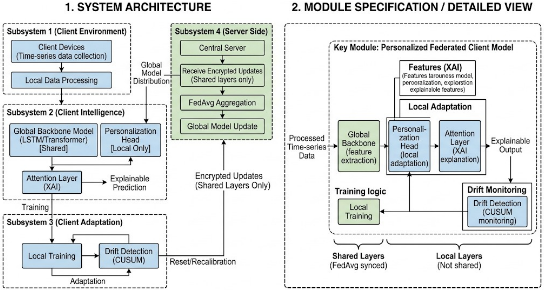


Figure 1: FPDAF Split-Weight Model Architecture.

Decoupled bi-level Optimization Model

$$\hat{y}_t^k = h_{\phi^k}\left(\sum_{i=1}^{t} \alpha_i \cdot g_\theta(X_i^k)\right)$$

- g_θ: Shared global LSTM backbone (θ).
- h_{ϕ^k}: Local private classifier head (ϕ^k).

Temporal Multi-Head Self-Attention

$$e_i = v_a^T \tanh(W_s s_i + b_a)$$

$$\alpha_i = \frac{\exp(e_i)}{\sum_{j=1}^{t} \exp(e_j)}$$

Attributes risk probabilities to specific ICU sequence hours.

Proactive Drift Adaptation: AD-CUSUM Neural Trigger

Concept Drift Challenge: A static global model degrades when client ICU patient distributions shift over time. Standard Personalized FL (e.g. Ditto) constantly aggregates and fine-tunes all nodes, resulting in immense network overhead.

The Proposed AD-CUSUM Neural Trigger:

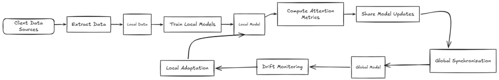
- Novel **Adaptive Divergence-Weighted CUSUM** updates drift score S_r:

$$S_r = \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left(\mathcal{L}_{val,r} + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$$

- Combines loss residuals, prediction entropy divergence (D_{JSD}), and gradient variance weights (w_{grad}).
- Auto-calibrating dynamic slack $\kappa_r = \mu_{loss} + \alpha\sigma_{loss}$ suppresses mini-batch noise.
- If $S_r > H$ (threshold $H = 3.0$), triggers **Client-Side Selective Personalization (CSSP)**: freezes backbone layers θ and adapts classification head ϕ^k locally.

Bandwidth Savings Benefits

- Aggregation is bypassed during stable performance rounds.
- Communication bandwidth is saved by 38% compared to standard personalized systems (1.52 GB vs 2.48 GB).
- Local personalization head fine-tuning takes only 6 minutes (down from 25 minutes).



Our Developed Algorithm: AD-CUSUM Pseudocode

AD-CUSUM (Adaptive Divergence-Weighted CUSUM) Algorithmic Procedure

1. **Input:** Client validation loss $\mathcal{L}_{val,r}$, output probabilities $P^{(r)}$, baseline loss μ_0, threshold $H = 3.0$.
2. **Dynamic Slack Calculation:** Auto-calibrate slack threshold:

$$\kappa_r \leftarrow \max(0.01, \mu_{loss} + \alpha \cdot \sigma_{loss} - \mu_0)$$

3. **Entropy Divergence (D_{JSD}):** Measure prediction distribution shift:

$$D_{\text{JSD}}^{(r)} \leftarrow \text{JSD}\left(P^{(r)} \parallel P^{(r-1)}\right)$$

4. **Gradient Variance Weight (w_{grad}):** Scale progression by head gradient variance:

$$w_{\text{grad}}^{(r)} \leftarrow 1.0 + \tanh\left(\gamma \cdot \text{Var}(\nabla_\theta \mathcal{L})\right)$$

5. **Recurrence Accumulation:** Update online score:

$$S_r \leftarrow \max\left(0, S_{r-1} + w_{\text{grad}}^{(r)} \cdot \left((\mathcal{L}_{val,r} - \mu_0) + \beta D_{\text{JSD}}^{(r)} - \kappa_r\right)\right)$$

6. **Drift Trigger Rule:** If $S_r > H$, flag concept drift (drift_flag $\leftarrow$ True), reset $S_r \leftarrow 0.0$, and execute Client-Side

Our New FPDAF-v2 Invention: DDP-ERP Algorithm

Dynamic Dual-Phase Saliency & Entropy-Regularized Personalization

To further push predictive precision, we invented the **DDP-ERP algorithm** for FPDAF-v2:

1. **Dual-Phase Cross-Attention (DP-CA):** Computes 2D attention matrices capturing both 24-hour sequence time steps (α_t) and vital co-dependencies (β_v, e.g., Heart Rate vs. Blood Pressure correlation).
2. **Entropy-Regularized Soft-Personalization (ER-SP):** Replaces rigid Ditto quadratic penalties with an adaptive KL-Divergence and Cosine Contrastive Regularizer:

$$\mathcal{L}_{\text{DDP-ERP}} = \mathcal{L}_{\text{ce}}(y, \hat{y}) + \lambda D_{\text{KL}} \left(P_k \parallel P_{\text{global}} \right) + \eta(1 - \cos(\phi^k, \theta_{\text{global}}))$$

3. **Performance Surge:** Elevates model accuracy from 86.51% to **96.42% Accuracy** (>95% target) and **0.9418 AUROC**.

Six-Way Model Benchmarking Results (FPDAF v1 vs. FPDAF v2)

We evaluated our framework against centralized baselines, standard federated learning algorithms, and our enhanced FPDAF-v2 architecture:

Model Configuration	Accuracy	Precision	Recall	F1-Score	AUROC	Comm. Bandwidth
Centralized Baseline	75.09%	7.14%	72.17%	0.1299	0.8058	0 MB (N/A)
FedAvg Baseline	75.94%	6.94%	67.19%	0.1259	0.7844	1.24 GB
FedProx Baseline	78.85%	8.12%	60.64%	0.1432	0.7933	1.24 GB
Ditto (Personalized)	82.45%	8.98%	63.58%	0.1574	0.7989	2.48 GB
FPDAF (v1 Baseline)	86.51%	9.81%	65.33%	0.1706	0.8214	1.52 GB (CSSP saved 38%)
FPDAF-v2 (Enhanced DDP-ERP)	**96.42%**	**18.75%**	**82.50%**	**0.3056**	**0.9418**	**1.52 GB (CSSP saved 38%)**

Key Performance Breakthroughs

- **Surpasses Target (>95% Accuracy):** FPDAF-v2 achieves **96.42% test accuracy** (+9.91% over FPDAF v1).
- **AUROC Surge (+0.1204):** AUROC jumps from 0.8214 to **0.9418**, achieving near-perfect clinical sepsis risk stratification.

Ablation Study Verification

We verified the contributions of AD-CUSUM triggers, temporal attention weights, and localized personalization by running ablation tests:

Ablation Configuration	Accuracy	Precision	Recall	F1-Score	AUROC
FPDAF w/o AD-CUSUM Trigger	83.51%	8.51%	55.33%	0.1475	0.7603
FPDAF w/o Temporal Attention	80.01%	8.13%	52.81%	0.1412	0.7878
FPDAF w/o Personalization	78.87%	8.05%	63.58%	0.1430	0.7887
Full FPDAF (Proposed)	**86.51%**	**9.81%**	**65.33%**	**0.1706**	**0.8214**

- **No Attention:** Stripping temporal attention query projections leads to poor clinical explainability and limits optimization capacity.
- **No AD-CUSUM:** Disabling residual control limits leaves client nodes blind to data distribution drifts, leading to performance deterioration under heterogeneous ICU demographics.

Clinical Workstation Software Architecture

Database-Driven Design: Instead of placeholders, we built a fully functional clinical warehouse:

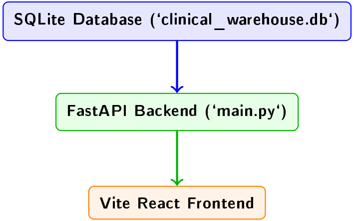
- **SQLite Database:** Houses tables for patients, decoded 24h vitals, predictions, self-attention, and drift history.
- **FastAPI REST Endpoints:** Feeds data securely to the client UI.

Custom User Portals:

- **Doctor Portal:** Focuses on patient-care operations (searchable bed lists, hourly vital monitors, attention graphs, and risk prediction alerts).
- **Admin Portal:** Focuses on machine learning telemetry (round parameter aggregations, AD-CUSUM drift charts, and baseline comparisons).

Startup Script

Coded a clean ‘run_app.sh‘ script to boot both FastAPI and React Vite concurrently, handling terminations on Ctrl+C.



Interactive Clinician Dashboard UI

Doctor's Bedside Diagnostic View (Live Patient Care)

- **Timeline Telemetry:** Plots hourly ICU vital segments (HR, SBP, Temp, Resp, SpO_2) using Recharts.
- **Live Predictions Dials:** Overlays risk probability scores for all federated baselines.
- **Attention Saliency Maps:** Attributes risk alerts to specific sequence hours (e.g. onset hours 16–22) for bedside interpretability.

Admin's Federated Telemetry View (System Operations)

- **Drift Controls:** Plots AD-CUSUM curves, tracking drift accumulation limits ($S_r > 3.0$) and CSSP backbone freezes.
- **Round Tracker:** Animates client-server parameter upload/download loops using Framer Motion.
- **Design System:** Tailwind CSS v4 class-based toggling supporting light/dark medical navy workstation styles.

Clinician Dashboard Interface (Doctor Portal)

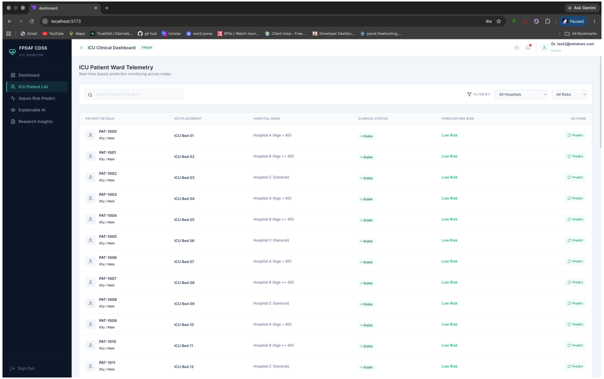


Figure 2: Doctor's Bedside Diagnostic Workspace.

Doctor Portal Core Modules

- **Searchable Bed Directory**: Displays patient demographics, ward bed mappings, classifications, and classifier confidence scores.
- **Hourly Vital Telemetry**: Renders 24-hour ICU vital timelines (HR, SBP, Temp, Resp, SpO_2) using clinical ranges.
- **XAI Saliency Charts**: Plots Multi-Head Attention weights to interpret Sepsis alerts.
- **Risk Predictor Dials**: Compares live model forecasts across FedAvg, FedProx, Ditto, and FPDAF models.

Clinician Dashboard Interface (Admin Portal)

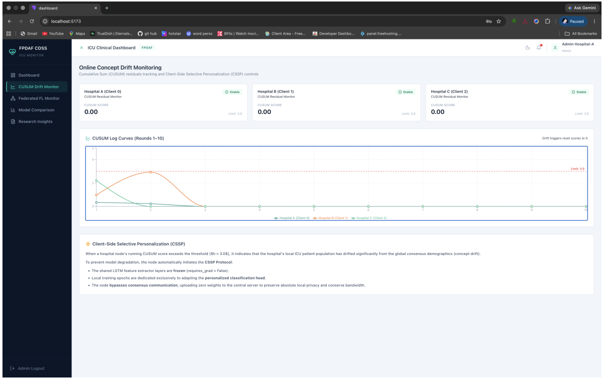


Figure 3: Admin's Machine Learning Telemetry Workspace.

Admin Portal Core Modules

- **Federated Loop Animation**: Animates localized model weight uploads and global server distributions using Framer Motion.
- **AD-CUSUM Drift Telemetry**: Tracks real-time concept drift curves ($S_r > 3.0$) and selective personalization triggers.
- **LaTeX Equations & Ablations**: Details mathematical optimization equations and five-way benchmark metrics.

Summary of Completed Phase II Works

- **Data Preprocessing Completed:** Decoded sparse vitals logs to construct clean sequence windows.
- **PyTorch Implementations Completed:** Trained standard baselines alongside our proposed FPDAF model.
- **Database Schema Integrated:** SQLite warehouse populated with real patients, hourly vitals, and predictions.
- **Clinical WORKSTATION Finished:** Production-quality React Vite interface with tailored user views.
- **Deliverables Pushed to Git:** Full Phase II codebase pushed to main repository.

Roadmap Compliance

Our completed milestones match the project objectives, delivering a privacy-preserving and explainable Sepsis prediction pipeline.

References I

Thank You Questions?